1. **Coulomb law:** Forces are vectors: the force between two charges in a 3d space has the same direction of the line connecting the two charges.

$$\boldsymbol{F} = k\frac{q_1 q_2}{r_{12}^2}\boldsymbol{u}_{r_{12}} = k\frac{q_1 q_2}{r_{12}^3}\boldsymbol{r}_{12}$$

$\mathbf{u}_{r12}$ is a unitary vector in the direction of $\mathbf{r}_{12}$

2. Electric field:

$$\boldsymbol{F} = q \cdot k\frac{Q}{r_{12}^2}\boldsymbol{u}_{r_{12}}$$

Suppose we keep a charge q fixed in a point in space, and we measure the force $\boldsymbol{F}$ that a second charge q feels, as it moves around the charge Q.

Since the force $\boldsymbol{F}$ is proportional to q, the ratio between $\boldsymbol{F}$ and $q2$ depends only on the specific position in space:

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{\boldsymbol{F}(\boldsymbol{r})}{q}$$

We call $\boldsymbol{E}(\boldsymbol{r})$ the electric field generated by Q and we can think at it as a perturbation of the empty space, due to the existence of Q.

3. Electrostatic potential

The potential energy of a charge q, which is interacting with another charge Q is proportional to q.

$$\boldsymbol{F} = k\frac{qQ}{r_{12}^2}\boldsymbol{u}_{r_{12}} \Rightarrow U = +k\frac{qQ}{r_{12}} = q\left(+k\frac{Q}{r_{12}}\right)$$

In general, the potential energy felt by a charge q is proportional to the charge itself.

Similarly to the definition of the electric field, we can now define the electrostatic potential as the ration between the electrostatic potential energy and the charge that feels that energy:

$$V(\boldsymbol{r}) = \frac{U(\boldsymbol{r})}{q} \qquad \boldsymbol{E} = -\boldsymbol{\nabla} \boldsymbol{V}$$

4. Polarization of atoms and molecules

Electrons are distributed in orbitals around the atomic nuclei.

When an external electrostatic field is applied, **the electron cloud shifts slightly relative to the nucleus.**

This creates **an electric dipole moment** which tends to align with the external field.

The electrostatic attraction between electrons and nuclei differs for different atoms.

As a result, in a molecule, electrons may be more localized around one atom than another, leading to **partial charges.**

Depending on the geometry, some molecules have **a permanent dipole moment**, even in the absence of external fields.

5. Solvation (hydration) shells

A solvation shell is the structured layer of solvent molecules surrounding a solute (e.g., an ion).

Because water is polar and ions are charged, **solvation shells are relatively stable and can mask the ion's charge.**

Multiple solvation shells may form, but they become progressively less ordered and less stable farther from the ion.

Solvation shells are dynamic, with solvent molecules constantly exchanging between the shells and the bulk.

6. Water dielectric constant

When an external electric field is applied, molecular dipoles (like water) tend to reorient in the direction of the field.

This collective reorientation creates an induced field that opposes the external field, partially canceling it.

The stronger the dipoles and their alignment, the greater the reduction of the effective field inside the medium.

The dielectric constant (ε_r) quantifies this effect: it measures how much a medium reduces the effective electrostatic interaction between charges compared to vacuum.

Water has a high dielectric constant (~80 at room temperature), meaning it strongly screens charges and stabilizes ions and polar molecules in solution.

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \rightarrow E = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{q}{r^2}$$

In vacuum In water

7. Wave functions

$\Psi(x)$: wave function. It represent the quantum state of the system. The function $\Psi(x)$ can be used to determine the probability that a particle (i.e. an electron, e) is found in a certain part of the space (i.e the interval [a,b]).

$$P(e \in [a,b]) = \int_a^b |\Psi(x)|^2 dx$$

probability density:

$$\rho(x) = |\Psi(x)|^2$$

8. Measurements and collapse of the wave function

When we take a measurement (for example the position of an electron), we do not get the wave function, but we get the result of the measurement (for example the coordinates x,y,z).

After the measurement, we know where the electron is, and it cannot be in all the other possible places.

A possible interpretation is that the measurement collapses the wave function!

The wave function changes due to the measurements.

9. The equation of Schroedinger

This equation describes the time evolution of the wave function:

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t) = \left[\frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\Psi(x,t)$$

Total energy of the system: $\mathrm{H} = \frac{1}{2}mv^2 + V = \frac{p^2}{2m} + V$

Time independent solutions of the equation of Shroedinger will obey the following property: $H\Psi(x) = E\Psi(x)$

10. Orbitals

$$HR(r)Y(\theta,\phi) = ER(r)Y(\theta,\phi)$$

The equation of Shroedinger has solution only for discrete values of E.

The energy of an electron bound to an atom is quantized.

The solutions give us the energy values, orbital “sizes” and orbital “shapes”.

For the hydrogen atom, the energy levels do not depend on the angular part of the solution.

But in general different orbital shapes have different energy.

Orbitals are characterized by few **quantum numbers** that corresponds to the possible solutions of the Schroedinger equation for that atom: Radial part of the orbital and Angular part of the orbital.

11. Spins and Pauli’s exclusion principle

In addition to the quantum numbers that describe an orbital's energy and shape, **electrons have an intrinsic quantum property called spin.**

This property is quantized and can have one of two values: spin-up (+ 1/2) or spin-down (− 1/2).

Pauli's exclusion principle states that **no two electrons in the same atom can have the same set of four quantum numbers.**

An atomic or molecular orbital can hold a maximum of two electrons, and these two electrons must have opposite

spins.

This explains why electron shells fill up in a specific order, which in turn determines the chemical properties of atoms and the nature of chemical bonds that form molecules.

Molecular orbitals: Energy for excitation: 10 eV

12. Classical Physics

we live at 310K, and biology uses only few *light* atoms, Thermal energy at room temperature: **26 meV,** Electronic energy levels: around **10 eV. The energy cost to excite light atoms is much higher than the thermal energy.**

Light atoms (and molecules made of them) are all in the "ground state" at room temperature.

Chemical reactions and heavy atoms are very well controlled within any living organism.

13. Molecules according to classical physics

Each atom is treated as a "classical" object interacting with the other atoms in the system.

The interaction is represented through the classical potential energy.

The equilibrium state is the minimum of the potential energy.

Different configurations are possible, without altering the electronic configurations of the molecules.

From the energy we can get the forces, and the time evolution.

At 310K different states are accessible, so Entropy exists, and Free Energy will decide the most probable state for the system.

14. Classical Force Fields

Since the electronic configuration is frozen in the ground state, we can simplify the form of the energy of interaction in many different ways:

1) A charge is assigned to each atom, and it is equal to the average charge of that atom.

2) Interactions between bonded atoms reproduce chemical and geometrical properties of the molecule, in particular regarding bond distances, angles and strength.

3) Non-bonded atoms interact exclusively through Coulomb and Lennard-Jones potentials.

15. Classical force field potential energy

$$U = \sum_{bonds} k_r(r - r_0)^2 + \sum_{angles} k_\theta(\theta - \theta_0)^2 + \sum_{dihedrals} C_k \cos(\varphi - \varphi_{0,k})) + \sum_{impropers} k_\alpha(\alpha - \alpha_0)^2 + \sum_{i<j} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} + \sum_{i<j} \left(\frac{A}{r_{ij}^{12}} - \frac{B}{r_{ij}^{6}}\right) + \sum_{h.b.} \left(\frac{C}{r_{ij}^{12}} - \frac{D}{r_{ij}^{10}}\right)$$

This expression is a classical force field potential energy written as a sum of bonded terms (bonds/angles/dihedrals/**Improper dihedrals**) and non-bonded terms (electrostatics/van der Waals/an explicit H-bond).

a) Bond stretching term

Deformation of chemical bond lengths away from their equilibrium value. "**Atoms oscillate around their equilibrium average distances; thus, chemical bonds are treated as harmonic oscillators.**"

r= bond length, r_0= equilibrium bond length, k_r= spring constant (bond stiffness).

b) Angle bending term

Deformation of bond angles away from equilibrium. Two consecutive bonds form an angle, we assume a **harmonic approximation**: the angle can oscillate around the equilibrium configuration with a well-defined angular strength.

θ= angle, θ_0= equilibrium angle, k_θ= angular stiffness.

c) Dihedral (torsional) term

The energetic preference for rotation about a bond, controlling conformations. "**Four consecutive atoms define three bonds and two planes. The two planes describe an angle which is called dihedral angle.**

φ: dihedral angle; C_k: strength 2for component k; $\varphi_{0,k}$: phase (preferred offset).

d) Improper dihedral term

Constraints that maintain planarity of specific atom groups. Carbon rings and other similar structure display planar configurations due to the specific electronic configurations. Improper dihedral angles need to be introduced to keep such atoms in the same plane.

α= improper dihedral, α_0= target value, k_α= stiffness.

e) Coulomb (Electrostatic) non-bonded term

Coulomb interactions between atom pairs that are treated as non-bonded. The charge represents the average charge on the atom due to the distribution of the electrons in that particular molecule.

q_i, q_j: atomic charges; r_{ij}: distance between atoms $i$and j; $i < j$avoids double counting, ε_0: vacuum permittivity.

f) van der Waals (Lennard–Jones) term (non-bonded)

Non-bonded short-range repulsion (A/r^{12}) and longer-range **attraction** (dispersion) (B/r^6) between atoms $i$and j. It is the sum of an attractive part (quantum effect) and a repulsive part which represent the effect of electronic clouds in close proximity.

A, B: parameters; r_{ij}: interatomic distance; $i < j$avoids double counting.

g) Hydrogen-bond term

An empirical term to represent hydrogen-bond interactions. Hydrogen bonds: an empirical potential that take explicitly into account of the possibility of creation of hydrogen bonds.

C/rij^{12}: short-range repulsion; $-D/rij^{10}$: stabilizing attraction.

16. The Lennard-Jones potential

At short distances, the LJ potential is strongly repulsive. Atoms that are too nearby in space create strong repulsions, that could be enough to completely distort the protein structure.

This happens very often when hydrogen atoms are added to the system (for example for a structure downloaded from the PDB or after a docking).

Energy has to be carefully minimized before starting MD.

17. RNA secondary and tertiary structure

RNA is mostly a single strand molecule and can form nucleotide pairs with itself. Contrarily to DNA, RNA can also have structural roles.

18. Side-chain (R-group) properties

Electrically charged side chains

Positively charged (basic): Arginine (Arg, R), Histidine (His, H), Lysine (Lys, K)

Negatively charged (acidic): Aspartic acid (Asp, D), Glutamic acid (Glu, E)

These residues form electrostatic interactions (salt bridges) and contribute to pH-dependent behavior.

Polar, uncharged side chains

Serine (Ser, S), Threonine (Thr, T), Asparagine (Asn, N), Glutamine (Gln, Q)

These residues commonly form hydrogen bonds and are often found on protein surfaces or in active sites.

Hydrophobic side chains

Alanine (Ala, A), Valine (Val, V), Isoleucine (Ile, I), Leucine (Leu, L), Methionine (Met, M), Phenylalanine (Phe, F), Tyrosine (Tyr, Y), Tryptophan (Trp, W)

These residues contribute to the hydrophobic effect and are frequently enriched in the hydrophobic core of folded proteins.

Cysteine (Cys, C): contains a **thiol (–SH)** group; can form **disulfide bonds (Cys–Cys)**, which stabilize protein structure.

two Cys can form a **disulfide bond (–S–S–)** under oxidizing conditions, which **stabilizes the folded structure** by covalently linking parts of the polypeptide.

Glycine (Gly, G): smallest side chain (**H only**); gives **high flexibility** and can fit into tight turns, allowing (ϕ, ψ)region broader, it has less steric hindrance, allowing more backbone conformations.

Proline (Pro, P): side chain bonds to the backbone **nitrogen** forming a rigid ring; **strongly restricts ϕ** and lacks an amide H for backbone H-bonding, so it often **breaks or kinks α-helices** and introduces turns/bends.

19. Ramachandran Plots

The relative position of two consecutive amino acids is described by two dihedral angles (φ and ψ).

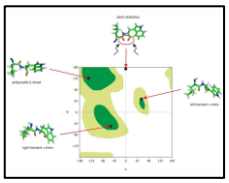
Not every combination of φ and ψ is possible, due to steric and electrostatic hindrance.

The values of the two dihedral angles is linked to the local secondary structure.

Figure shows the typical protein Ramachandran plots.

a) Provide a brief explanation of what the two angles (φ and ψ) in the plot are.

b) Write a short caption for Figure, explaining in particular what the light green and dark green regions represent.



a) The relative position of two consecutive amino acids in a polypeptide backbone is described by two dihedral (torsion) angles, ϕ and ψ.

ϕ (phi) is the backbone dihedral angle around the N–Cα bond.

ψ (psi) is the backbone dihedral angle around the Cα–C bond.

Not every (ϕ, ψ) combination is possible because of steric and electrostatic hindrance.

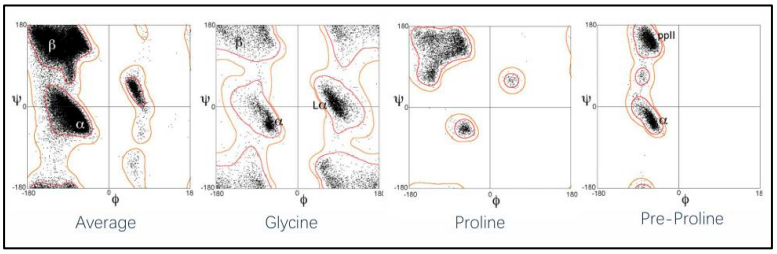
The values of ϕ and ψ are linked to the local secondary structure.

b) Ramachandran plot showing the distribution of backbone dihedral angles (ϕ, ψ) for amino-acid residues. The shaded regions represent (ϕ, ψ) combinations that are sterically/energetically allowed, while unshaded regions are disallowed due to steric and electrostatic hindrance. Dark green indicates the most favoured (most probable) conformational regions; light green indicates additional allowed but less favoured regions. Clusters correspond to common secondary structures such as the right-handed α-helix region (typical ϕ ≈ −60°, ψ ≈ −45°) and β-sheet regions.

Different amino acids have different preferred orientation (which explain their secondary structure propensity).

a) Identify which amino acid is proline and which is glycine and justify your answer.

b) Explain what the regions in the plot marked with α and β represent.



The left Ramachandran plot corresponds to glycine, and the right plot corresponds to proline.

a) Average

This is the "typical" Ramachandran distribution for most residues: it shows the main allowed regions that correspond to common secondary-structure conformations.

b) Glycine

Compared with "Average", glycine shows a much broader allowed area (more ϕ/ψ combinations are populated).

Glycine has the smallest side chain (H), so it has less steric hindrance, allowing more backbone conformations. Different amino acids have different preferred orientations.

c) Proline

Proline shows a much more restricted distribution than "Average".

The proline side chain forms a ring with the backbone, which strongly constrains backbone geometry, **particularly the ϕ angle, so fewer (ϕ, ψ)combinations are sterically allowed.**

d) Pre-Proline

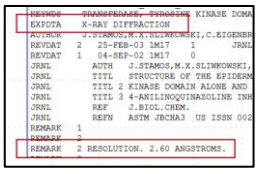
This plot corresponds to residues immediately preceding a proline in sequence, and it shows a distinct preference pattern compared with "Average".

The labeled pPII region indicates a polyproline-II-type backbone conformation that is commonly enriched when the next residue is proline (the neighboring proline imposes steric constraints that shift the preferred ϕ/ψ space).

Because **ϕ and ψ are linked to the local secondary structure**, the labeled regions correspond to common backbone conformations: **α region**: backbone dihedral combinations typical of an **α-helix** (right-handed). In the lecture table, α-helix angles are approximately **ϕ ≈ −60°** and **ψ ≈ −45°**.

β region: backbone dihedral combinations typical of **β-strands/β-sheets**

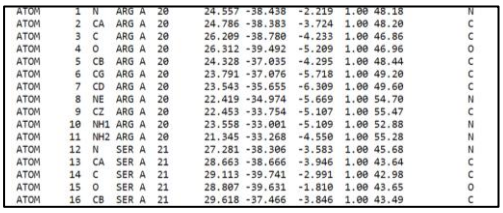
20. The Protein Data Bank and PDB files



```
KEYWDS    TRANSFERASE, TYROSINE KINASE DOMA
EXPDTA    X-RAY DIFFRACTION
AUTHOR    J.STAMOS,M.X.SLIWKOWSKI,C.EIGENBR
REVDAT   2   25-FEB-03 1M17    1       JRNL
REVDAT   1   04-SEP-02 1M17    0
JRNL        AUTH   J.STAMOS,M.X.SLIWKOWSKI,
JRNL        TITL   STRUCTURE OF THE EPIDERM
JRNL        TITL 2 KINASE DOMAIN ALONE AND
JRNL        TITL 3 4-ANILINOQUINAZOLINE INH
JRNL        REF    J.BIOL.CHEM.
JRNL        REFN   ASTM JBCHA3  US ISSN 002
REMARK   1
REMARK   2
REMARK   2 RESOLUTION. 2.60 ANGSTROMS.
```

EXPDTA: The **experimental method** used to determine the structure.

REMARK 2: RESOLUTION (Å): The **resolution** of the experimental data.



```
ATOM      1  N   ARG A  20      24.557 -38.438  -2.219  1.00 48.18           N
ATOM      2  CA  ARG A  20      24.786 -38.383  -3.724  1.00 48.20           C
ATOM      3  C   ARG A  20      26.209 -38.780  -4.233  1.00 46.86           C
ATOM      4  O   ARG A  20      26.312 -39.492  -5.209  1.00 46.96           O
ATOM      5  CB  ARG A  20      24.328 -37.035  -4.295  1.00 48.44           C
ATOM      6  CG  ARG A  20      23.791 -37.076  -5.718  1.00 49.20           C
ATOM      7  CD  ARG A  20      23.543 -35.655  -6.309  1.00 49.60           C
ATOM      8  NE  ARG A  20      22.419 -34.974  -5.669  1.00 54.70           N
ATOM      9  CZ  ARG A  20      22.453 -33.754  -5.107  1.00 55.47           C
ATOM     10  NH1 ARG A  20      23.558 -33.001  -5.109  1.00 52.88           N
ATOM     11  NH2 ARG A  20      21.345 -33.268  -4.550  1.00 55.28           N
ATOM     12  N   SER A  21      27.281 -38.306  -3.583  1.00 45.68           N
ATOM     13  CA  SER A  21      28.663 -38.666  -3.946  1.00 43.64           C
ATOM     14  C   SER A  21      29.113 -39.741  -2.991  1.00 42.98           C
ATOM     15  O   SER A  21      28.807 -39.631  -1.810  1.00 43.65           O
ATOM     16  CB  SER A  21      29.618 -37.466  -3.846  1.00 43.49           C
```

a) Everything following the 'ATOM' word indicates the properties of an atom in the protein.

Record type = ATOM (this line describes an atom record in a PDB file).

Atom number (atom serial index in the file).

Atom "Type"/Atom name (e.g., N, CA, C, O, CB, CG…).

Residue name (3-letter code).

Chain identifier (distinguishes different polymer chains).

Residue number (sequence index of the residue in that chain).

x coordinate (in Å).

y coordinate (in Å).

z coordinate (in Å).

Element (chemical element symbol, e.g., C, N, O).

Occupancy (fraction of time spent in that position for each atom).

B-factors (fluctuations around the average positions Å²).

b) Because a residue contains multiple distinct carbon atoms, the PDB format must assign unique atom names so each atom can be unambiguously identified in the structure.

In amino acids, the backbone carbons have standard names:

CA = α-carbon (the central carbon of the amino acid backbone)

C = carbonyl carbon of the peptide backbone

The side-chain carbons are named by position away from CA: CB, CG, CD …

Some residues have functional group specific naming, e.g., ARG, because these are chemically different carbons in the side chain.

So different carbon atoms must be labeled differently

21. Anfinsen used ribonuclease A (RNase A) to study protein folding.

Treatment with 8 M urea and β-mercaptoethanol produces a denatured, reduced RNase A: the protein is unfolded and its disulfide bonds are broken, so it loses the native, catalytically active conformation.

β-mercaptoethanol is a reducing agent that **cleaves disulfide bonds** by reducing **S–S** linkages back to **two thiols (–SH)** on cysteine residues.

Urea is a denaturant that disrupts stabilizing interactions in proteins, leading to **unfolding of tertiary and secondary structure** (exposing internal regions to solvent).

Anfinsen's experiment shows that the **native conformation** of RNase A (with the **correct disulfide bonds**) can be recovered after complete denaturation and reduction, provided the denaturant is removed to allow refolding before oxidation. This supports the conclusion that the information required to reach the **native, catalytically active** structure is contained in the **amino-acid sequence**, and that the native state corresponds to a thermodynamically favored structure under the right conditions; conversely, oxidation while unfolded leads to **scrambled disulfides** and loss of activity.

22. Levinthal's paradox and funnel

Protein cannot explore the whole configuration space to find the native state (minimum of energy).

A possible way to solve Levinthal's paradox is assuming that the energy landscape is shaped as a funnel.

The funnel shape represents a bias toward the native state, with energy decreasing as the protein approaches its correctly folded structure.

Multiple folding pathways can lead to the same native state, eliminating the need to search through all possible conformations randomly.

The width of the funnel represents conformational entropy (number of possible states), which decreases as folding progresses and the structure becomes more ordered.

Local minima in the funnel can act as folding intermediates or misfolded traps, but the overall downward slope drives folding toward the global energy minimum.

Folding is primarily driven by the hydrophobic effect, where hydrophobic residues collapse into the protein core to minimize unfavorable contacts with water, while hydrophilic residues remain on the surface.

23. Homology Modeling

Homology modeling, also known as comparative modeling, uses homologous sequences with known 3D structures for the modeling and prediction of the structure of a target sequence

Basic assumption: two homologous proteins will share very similar structures

Although the number of actual proteins is vast, there is a limited set of tertiary structural motifs to which most proteins belong

Homology modeling is one of the best-performing methods and (in most cases) gives accurate predicted models

24. *Ab initio* modeling

Ab initio: (Latin) from the beginning.

Ab initio methods aim to predict protein structures from their amino acid sequences without relying on known structural templates.

These methods often use physics-based energy functions or machine learning techniques to sample the conformational space and identify low-energy structures.

These methods are very time-consuming since the configuration space is incredibly large.

Very interesting from a theoretical point of view, but do not produce reliable results yet.

25. AlphaFold

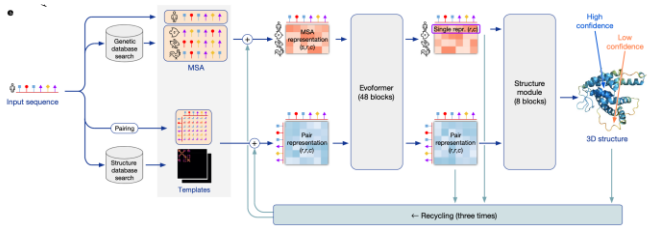
AlphaFold utilizes deep learning and advanced machine learning techniques to accurately predict protein structures.

The key innovations of AlphaFold include:

Attention-based Neural Network: AlphaFold uses an attention-based neural network architecture that can capture long-range interactions between amino acids.

Evolutionary Information: AlphaFold incorporates evolutionary information from multiple sequence alignments.

End-to-End Approach: AlphaFold is an end-to-end approach that directly predicts the 3D structure from the amino acid sequence without relying on intermediate steps or templates.



Input sequence

→ genetic database search → MSA

→ (optional) pairing

→ structure database search → templates

→ MSA representation + pair representation

→ Evoformer (48 blocks) updates representations

→ single (per-residue) repr. + refined pair repr.

→ Structure module (8 blocks)

→ 3D structure + confidence (high/low)

↺ Recycling ×3 (feed output back to refine)

26. Molecular Docking

Protein-Ligand interactions

Ligands are usually small and more rigid than proteins (few degrees of freedom).

Small molecules usually bind to pockets (cavities).

Interactions are often mediated by hydrophobic contributions.

In many practical cases, a large number of ligands need to be tested (screened).

Protein-Protein interactions

Proteins are large and can have important configuration changes upon binding.

The interaction between the proteins happens on their surface.

It is often mediated by hydrophilic residues and by surface complementarity.

The interacting proteins are usually known (no need for screening).

27. Binding affinity

a) The binding free energy of a ligand (small molecule A) to a protein can be described by the formula ΔG=ΔH−TΔS. Explain what ΔG, ΔH and ΔS are in this context and how they can be interpreted at the microscopic level.

b) Assume there is a second ligand (small molecule B) with binding affinity ΔGB<ΔGA. Explain how the two molecules will compete for binding to the protein if they are added to the solution at the same concentration.

c) Which of the two molecules in question (b) is a better drug candidate if we consider the binding affinity alone? Motivate your answer.

a) ΔG(binding free energy): the difference of free energy between the bound and unbound state,

$$\Delta G = G_B - G_U$$

and it can be written as

$$\Delta G = \Delta H - T\Delta S.$$

ΔH (enthalpy change upon binding): energy changes associated with breaking and forming intermolecular interactions (hydrogen bonds, van der Waals interactions, electrostatic interactions) during binding; it also includes (usually unfavorable) energy changes due to loss of solvation of the protein and ligand.

Microscopic interpretation: these are changes in the interaction (potential) energy between atoms/molecules.

ΔS(entropy change upon binding): change in the configurational space accessible to the proteins and ligands upon binding (usually unfavorable), plus release or reorganization of water molecules from the binding interfaces (can contribute favorably).

Microscopic interpretation: at physiological temperature multiple states/configurations are accessible ("Entropy exists"), and free energy determines the most probable state.

b) Comparing two ligands:

$$\Delta\Delta G = \Delta G_B - \Delta G_A.$$

If $\Delta G_B < \Delta G_A$, then $\Delta\Delta G < 0$, so B is a better binder than A (lower binding energy, better binding affinity).

Therefore, when A and B are added at the same concentration, B will preferentially occupy the protein binding sites (and will tend to displace A at equilibrium).

c) Ligand B, because $\Delta G_B < \Delta G_A$ implies better binding affinity (better binder).

Equivalently: stronger binding corresponds to smaller K_D(and larger K_A, since $K_A = 1/K_D$).

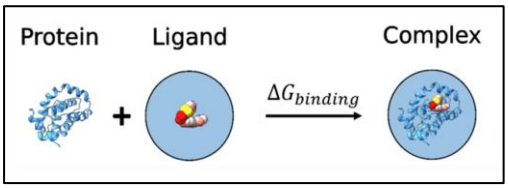
28. Binding affinity

Figure schematically represents the binding process of a molecule to a protein.

a) **The Gibbs Free Energy is given by: $G = U + pV - TS$. Explain what the internal energy $U$and the entropy $S$are from the microscopic point of view.**

b) **Define the binding Free Energy and explain what the difference in free energies between the bound and unbound states represents thermodynamically. Briefly explain what a negative value for ΔGmeans.**

c) **Describe how molecular simulations can be used to evaluate internal energy and entropy contributions to binding free energy.**



a) **Internal energy U**: microscopically, $U$is the system's energy of interaction between atoms/molecules, i.e., the (classical) potential energy describing how atoms interact.

Entropy S: microscopically, Sreflects the number of accessible microscopic states/configurations (configurational space).

b) **ΔG:** The binding free energy is the difference of free energy between the bound and unbound state.

$\Delta G_{binding}$=GB−GU

Thermodynamic meaning: The quantity $G_B - G_U$represents how much free energy changes upon forming the complex.

Negative ΔG: $\Delta G < 0$means binding is thermodynamically favourable, i.e., the bound state has lower free energy than the unbound state. This is consistent with the course link between **$\Delta G$and K_D: $\Delta G = RT\ln K_D$.**

c) **ΔG=GB−GU=UB−UU−T(SB−SU)**

Estimate the internal energy contribution from the interaction energy of the system.

Estimate the entropy contribution from how the simulation samples the configurational space accessible to protein/ligand.

29. Molecular Docking:

Autodock

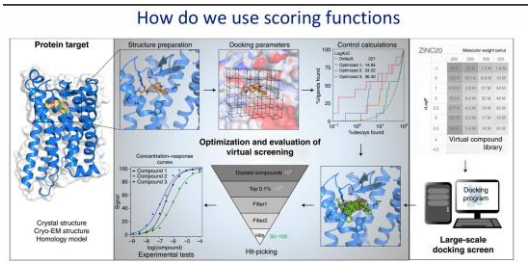
Autodock potential is based on Physics, but it introduce weighting pre-factor to each term (W) to better fit empirical data.

Schoedinger and Glide

Schoedinger is a software that use an empirical scoring function to determine the best mode and evaluate the binding energy of the docking.

30. How do we use scoring functions

31.



Input 1: Protein target

→ structure preparation

→ set docking parameters

Input 2: Virtual compound library

→ library preparation

Conbine

→ large-scale docking screen with a docking program

→ control calculations / retrospective validation (actives vs decoys; tune parameters; evaluate e.g., ROC/LogAUC)

→ rank docked compounds by scoring function

→ filtering/selection (top fraction → successive filters)

→ hit picking (e.g., ~50–100 hits)

→ experimental tests (concentration–response curves)

32. Protein–Protein: HADDOCK, AF3 accurately predicts structures across biomolecular complexes